Roll No.: DL.AI.U4AI0241140

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech. AI & Data Science Mid-Term Examinations – October 2024

First Semester

19ENG111 Technical Communication

Duration: Two hours | Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	To gain knowledge about the mechanics of writing and the elements of formal correspondence
CO02	To understand and summarise technical documents
CO03	To apply the basic elements of language in formal correspondence
CO04	To interpret and analyze information and to organize ideas in a logical and coherent manner
CO05	To compose project reports/ documents, revise them for language accuracy and make technical presentations

Multiple Choice: Answer all questions (10 marks)

1) What is the correct definition of an **Interrogative Statement**? [1] [CO1] [BT1]
   a. Those which make statements or assertions.
   b. Those which ask questions.
   c. Those which express commands, requests, or entreaties.
   d. Those which express strong feelings.
2) What is a **Literature Review**? [1] [CO5] [BT1]
   a. A critique of a set of publications based on standard evaluation criteria.
   b. A list of your favourite research articles.
   c. A systematic summary of the most relevant past publications to support your research.
   d. A historical chronicle of events tied to your research.
3) What does the word **Liminal** mean? [1] [CO1] [BT1]
   a. Occupying a position at a boundary or threshold; ambiguous
   b. Eager to take legal action to settle disputes
   c. Smelling or tasting of limes or citrus
   d. Lasting for only a short time; temporary
4) How many definitions are there of the word "Set"? [1] [CO1] [BT1]
   a. 3
   b. 13
   c. 130
   d. 430
5) If someone is described as "Incendiary," what kind of person are they? [1] [CO1] [BT2]
   a. Someone who is currently on fire.

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b. Someone who is extremely athletic.
c. Someone who is often confused or disoriented.
d. Someone who is confrontational and likely to cause a fight or argument.

6) What is the correct structure or order of a formal research argument? [1] [CO3] [BT1]
   a. Title, Abstract, Discussion, Conclusion
   b. Research Question, Thesis Statement, Literature Review, Data Analysis, Logical Tree, Counter Arguments, Conclusion
   c. Thesis Statement, Logical Tree, Literature Review, Counter Arguments, Data Analysis, Conclusion, Research Question
   d. Research Question, Thesis Statement, Logical Tree, Conclusion
7) What is NOT one of the goals of good research? [1] [CO4] [BT2]
   a. Refine and develop beliefs, assumptions, and principles of science.
   b. Prove/disprove critical theories.
   c. Inform and influence stakeholders.
   d. Increase personal and political influence.
8) When would you adopt a FORMAL or PROFESSIONAL tone in speaking or writing? [1] [CO3] [BT2]
   a. When speaking to your friends online.
   b. When passing someone on the street.
   c. When discussing your job roles with a supervisor.
   d. When asking an acquaintance for money.
9) Which of the following is NOT a part of every good paragraph? [1] [CO1] [BT1]
   a. Thesis question
   b. Topical sentence
   c. Logical Order
   d. Concluding sentence
10) Select all of the kinds of adjectives (more than one answer possible) [1] [CO1] [BT1]
   a. Quality & quantity
   b. Punitive
   c. Demonstrative
   d. Emphasizing

Short Answer: Answer all questions (20 marks)

1) **Write a self-introduction for a formal setting**. Include your name, where you are from, your role in Amrita University, and one fact about yourself. [5] [CO3] [BT6]
   a. **Answer**:

2) Correct the text. Re-write the following text so that it is correct: [5] [CO1] [BT3]
   a. **Original:** This mornin, a large strom hit the city, causing alot of damage to buildings and cars. Many peoples homes was flooded and powerlines falled down, leavin thousends without electricity? The govornor said in a statment that resque teams is workin hard to help everyone who has been affected by the strom. "We ask everyone to stay indoors until the water goes

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down," he sayed. Resedents are being advised to evacuate areas that are closest to the river, wich has overflown its banks. Weather experts warned that more rains is expected in the next few days.

b. **Your Corrected Text:**

3) Correct the text. Re-write the following text so that it is correct: [5] [CO3] [BT3]

a. **Original: *An email to your boss or supervisor***

Subject: Hey, quick question! :)

Hey [Boss's First Name],

Hope you're doing good! So, I was just wondering about that project we're working on. Do you think we can push the deadline a bit? I've been super busy with a bunch of stuff, and it's kinda hard to get everything done by Friday. No biggie if not, just thought I'd ask! :)

Also, any updates my salary increase? Would be cool to know what's going on with that.

Thanks a ton! Let me know!

Cheers,

[Your First Name]

b. Your Corrected Text:

4) Explain why it is important to Reference or Cite your work when writing a research paper? [5] [CO5] [BT2]

a. **Answer:**

Reading Comprehension: Read the following text and answer all questions (20 marks)

***See attached reference file:* "India turns to AI as cyber warfare threats grow.pdf"**

1) What is this article about? What are the main points? [4] [CO2] [BT4]

a. **Answer:**

2) According to the article, what are the problems India is facing in cyber security? [4] [CO2] [BT5]

a. **Answer:**

3) What are some of the solutions proposed by Innefu? [4] [CO2] [BT4]

a. **Answer:**

4) According to Jiten Jain, why has India been slow to adopt new Cyber Security methods? [4] [CO5] [BT5]

a. **Answer:**

5) What do you think? How can India improve its efforts to increase Cyber Security? [4] [CO4] [BT6]

a. **Answer:**

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks
CO01	11
CO02	12
CO03	12
CO04	5
CO05	10
TOTAL	**50**

BTL	Marks
BTL 1	7
BTL 2	8
BTL 3	10
BTL 4	8
BTL 5	8
BTL 6	9
TOTAL	**50**

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India turns to AI as cyber warfare threats grow

January 22 2017, by Megha Bahree

In the darkened offices of a tech start-up, a handful of computer engineers sifts through a mountain of intelligence data that would normally be the work of a small army of Indian security agents.

"We use artificial intelligence (AI) to look for patterns in the past to predict future behaviour," says Tarun Wig as he explains why he hopes his company *Innefu* can do more business with India's government. "Cyber warfare isn't a movie, it's happening right now.... We lost out on the industrial revolution, we lost out on the defence revolution—let's not lose out in the cyber revolution."

While other countries have long relied on AI to gather intelligence, India—sometimes seemingly addicted to paperwork—has continued to use agents to eyeball reams of data gathered over the years. It's a process that sucks up time and can often miss crucial information.

India has been in three wars with its neighbours since independence and the target of numerous cross-border attacks, including in 2008 when Pakistan-based extremists killed more than 160 people in Mumbai. Now the threat from cyber-attacks is growing and its vulnerability has been exposed.

Some 22,000 pages of data related to submarines that a French government-owned company was building for the Indian navy were leaked to the media last year. Opposition leader Rahul Gandhi's Twitter account was hacked in November while the elite National Security Guard's website was reportedly defaced with profanity-laden messages for Prime Minister Narendra Modi last month.

"Our idea starting out was that if the next war is fought on cyber, we need our own weapons," said Wig as he talks through software developed for India's needs.

Octopus tentacles

Innefu got a foot in the lucrative business of government contracts after resolving a thorny test case for a law enforcement agency that wanted to determine the background to an incident along one of India's borders.

The agency handed over two CDs with about 1,500 intelligence documents, including social media snippets, such as posts on planned protests. Innefu had to train the machine to read the agency's language, including abbreviations, and then began extracting information on what happened, who were the main players and how they interacted with each other.

Its newest offering *Prophecy* is modelled on products made by Palantir Technologies, a private security firm whose founders include Paypal cofounder Peter Thiel and whose clientele includes the CIA and the FBI. "Prophecy is like an octopus with multiple tentacles that pulls data from multiple places," said Wig's co-founder Abhishek Sharma.

While the use of AI is commonplace elsewhere in Asia, it is still in its infancy in India. About 75 percent of respondents to a recent survey by consulting firm EY India said cybersecurity deployed

in their organisations does not meet their needs, pointing to big opportunities for companies such as Innefu.

Although Innefu is the only Indian company known to specialise in national security, other Indian companies such as Arya.ai and Haptik are also tapping what should be a lucrative market. Banks have started to use AI to target products to customers and doctors are using it in a couple of experiments to map a patient's medical history in order to devise new lines of treatment. Work is also underway to create a system that can act as a backbone for all electronic warfare programmes for the Indian army. But most of this is still at the laboratory stage, experts say.

AI laggards

"It's expensive and our society is not used to automated decision-making," Jiten Jain, chief of Indian Infosec Consortium, an industry body, told AFP. "We're still used to manual and human decision making."

Subimal Bhattacharjee, a cyber security expert, said India had been caught off guard by the need to upscale its use of AI. "We are definitely laggards in comparison to China and South Korea and the US," he said.

Innefu operates from the eighth floor of a nondescript Delhi commercial complex that houses chartered accountants and eateries. Its windows are darkened with thick black paper, while clusters of wires sprout from walls adorned with photos of goddess Lakshmi, the purveyor of wealth. But despite its humble surroundings, it is confident an ongoing trial of Prophecy will lead to more business with India's security services—whom Wig expects to be naturally inclined to go local. "You can't really trust such sensitive data with foreign companies," said Wig.